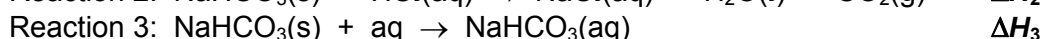
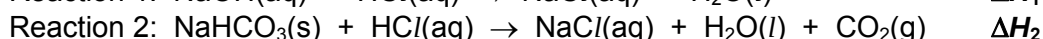


1 Planning (P)

The enthalpy change of reaction between carbon dioxide gas and aqueous sodium hydroxide, ΔH_r , cannot be determined directly by experiment. It is determined via an indirect approach using Hess' Law.



The enthalpy change of the following three chemical reactions can be determined directly by experiment.

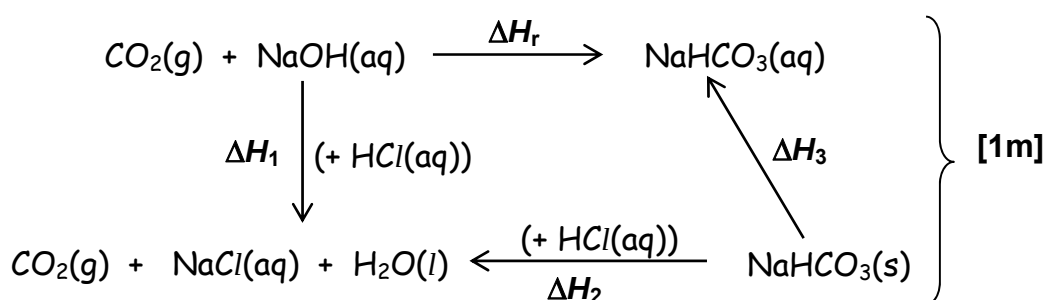


Using these enthalpy values obtained, ΔH_r can be determined using Hess' Law.

In a preliminary examination,

- ΔH_1 was found to be approximately -57 kJ mol^{-1} .
- ΔH_2 was found to be approximately $+27 \text{ kJ mol}^{-1}$.
- ΔH_3 was found to be approximately $+18 \text{ kJ mol}^{-1}$.

- (a) Construct an energy cycle to determine a value of ΔH_r , using the approximate values of ΔH_1 , ΔH_2 and ΔH_3 obtained in the preliminary examination.



$$\begin{aligned} \text{By Hess' Law, } \Delta H_r &= \Delta H_1 - \Delta H_2 + \Delta H_3 \\ &= (-57) - (+27) + (+18) \\ &= \underline{-66 \text{ kJ mol}^{-1}} \quad [1\text{m}] \end{aligned}$$

You are to design an experiment to verify the preliminary value of ΔH_2 given.

You are provided with the following information and a list of apparatus and reagents. You may also use other apparatus normally found in a school laboratory.

Reagents and apparatus:

- solid sodium hydrogen carbonate
- 2 mol dm^{-3} dilute hydrochloric acid
- a thermometer with 0.2°C division
- a stopwatch
- a 200 cm^3 styrofoam cup with lid (supported in a dry beaker)

Assume that 4.2 J is required to raise the temperature of 1 cm^3 of any solution by 1°C .

- (b) By considering the capacity of the styrofoam cup provided, state a suitable volume of hydrochloric acid you could use in the experiment.

Hence determine the mass of solid sodium hydrogen carbonate you should use in order to obtain an approximate temperature fall of 6 °C for reaction 2.

With your suggested quantities of chemicals used, prove that hydrochloric acid is used in excess in the experiment.

Volume of hydrochloric acid = 80 cm³ (30-50% capacity)

$$\begin{aligned}\text{Expected heat absorbed} &= 80 \times 4.2 \times 6 \text{ J} \\ &= 2016 \text{ J} = \underline{2.016 \text{ kJ}}\end{aligned}$$

$$\Delta H_2 \approx +27 \text{ kJ mol}^{-1}$$

$$+ \frac{2.016}{\left(\frac{m}{84.0}\right)} = +27$$

$$\begin{aligned}\text{Mass of solid NaHCO}_3, m &= \left(\frac{2.016}{27}\right) \times 84.0 \\ &= \underline{6.27 \text{ g}} \text{ (3 s.f. and units)}\end{aligned}$$

Correctly determine the mass of NaHCO₃ from the stated V_{HCl}
[1m]

$$\text{Amount of HCl(aq) used} = \frac{80}{1000} \times 2 = \underline{0.160 \text{ mol}}$$

$$\text{Amount of NaHCO}_3 \text{ used} = \frac{6.27}{84.0} = \underline{0.0746 \text{ mol}}$$

Since HCl ≡ NaHCO₃,

NaHCO₃ is limiting and HCl(aq) is used in excess.

Correctly calculate amount of both reactants and show that HCl is in excess using the mole ratio.
[1m]

- (c) Considering your answer in part (b), write a detailed plan to verify the enthalpy change of reaction between solid sodium hydrogen carbonate and hydrochloric acid, ΔH₂.

Your plan should include:

- all essential experimental details, including precision of apparatus and quantities of chemicals used.
- a sketch of graph you expect to obtain from your experimental data to correct for heat lost to/heat gained from the surroundings.
- an outline of how you would use the experimental results to calculate ΔH₂.

In your calculations, you should adhere to the following definition of terms.

Let *m* g be the actual mass of solid sodium hydrogen carbonate used;
T °C be the maximum temperature change.

You must simplify the mathematical expression in your final answer.

1. Weigh an empty weighing bottle using an electronic weighing balance which is precise to 2 decimal places.
2. Weigh accurately *6.27 g of solid NaHCO₃ into the weighing bottle and record the total mass.
3. Using a 100 cm³ measuring cylinder, add *80.0 cm³ of hydrochloric acid into the styrofoam cup.
4. Stir the solution in the cup using the 0.2 °C interval thermometer and record the initial temperature of the solution.
5. At a convenient time, start the stopwatch. Record the temperature of the solution in the styrofoam cup at 30 s intervals until 2.5 min.
6. At exactly 3 min, tip the weighed NaHCO₃ into the styrofoam cup.
7. Stir the contents in the styrofoam cup with the thermometer and record the temperature of the solution at 3.5 min. Continue to stir the contents in the styrofoam cup and record the temperature of the solution at 30 s intervals until 10.0 min.
8. Re-weigh the weighing bottle which contains some residual NaHCO₃ to find out the actual mass NaHCO₃ added to the styrofoam cup.

[1m] Weighing of *6.27 g of NaHCO₃ into a weighing bottle and reweighing the weighing bottle after tipping.

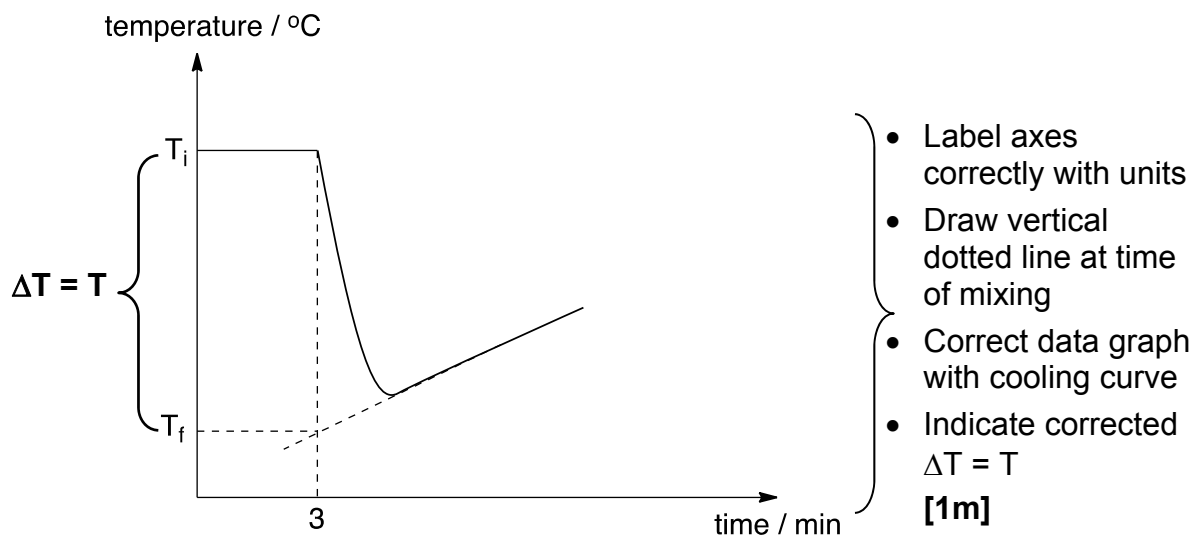
*mass value stated must be from the calculated mass in (b)

[1m] *Volume measurement of HCl(aq) using an appropriate accurate apparatus and measure initial temperature (of acid) inside the cup .

*volume of HCl measured must be from the stated volume in (b)

[1m] State time of mixing and take temperature measurements at *fixed time-intervals all the way to an appropriate end-time .

Expected graph:



Calculations to determine ΔH_2 :

$$\begin{aligned} \text{Heat absorbed} &= (80.0 \times 4.2 \times T) \text{ J} \\ &= (336T) \text{ J} = (0.336T) \text{ kJ} \\ \Delta H_2 &= + \frac{0.336T}{\left(\frac{m}{84.0}\right)} = +28.2 \left(\frac{T}{m}\right) \text{ kJ mol}^{-1} \end{aligned} \quad \left. \vphantom{\frac{0.336T}{\left(\frac{m}{84.0}\right)}} \right\} [1\text{m}]$$

- (d) Concentrated hydrochloric acid is a corrosive liquid with an approximate concentration of 11 mol dm^{-3} .

Describe clearly how you would prepare 250 cm^3 of 2 mol dm^{-3} of dilute hydrochloric acid from the given concentrated acid.

State all the necessary precautions you would take to dilute the acid.

$$\begin{aligned} M_1 V_1 &= M_2 V_2 \\ 11 \times V_1 &= 2 \times 250 \\ V_1 &= 45.5 \approx 46 \text{ cm}^3 \end{aligned}$$

Measure out 46.0 (or 45.0 or 45.5) cm^3 of concentrated HCl using a 50 cm^3 measuring cylinder.

Using another measuring cylinder, add 204.0 (or 205.0 or 204.5) cm^3 of deionised water into a 500 cm^3 beaker.

Then carefully pour the concentrated acid into the beaker containing the water. Stir with a glass rod to obtain a homogeneous diluted solution.

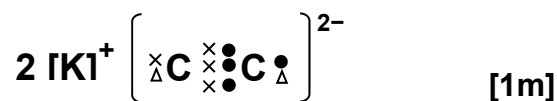
To prevent/avoid direct contact with the corrosive chemical, wear safety goggles, hand gloves and a laboratory coat during the preparation. Use a long glass dropper to add chemical while adjusting the solution level.

As corrosive/toxic/harmful acid fumes are produced, prepare the solution inside the fume-cupboard.

[Total: 12]

- 2 Carbides are excellent candidates for coatings for drills and other tools, as they are extremely hard and resistant to wear, corrosion and heat. Potassium and calcium carbides are commonly used in key industrial applications.

(a) (i) Draw the dot-and-cross diagram of potassium carbide, K_2C_2 .



(ii) State the type of hybrid orbitals around the carbon atoms in the carbide anion.

sp [1m]

(iii) Explain whether the melting point of K_2C_2 is expected to be higher or lower than that of CaC_2 .

Ca^{2+} has a higher charge and smaller radius than K^+ .

Since lattice energy, $LE \propto \frac{q_+ \times q_-}{r_+ + r_-}$, CaC_2 has a greater magnitude of lattice energy than K_2C_2 . Hence more energy is required to overcome the stronger electrostatic attraction (or ionic bonds) between Ca^{2+} and C_2^{2-} ions in CaC_2 than those between K^+ and C_2^{2-} ions in K_2C_2 . [1m]

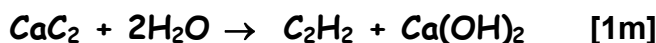
Therefore, K_2C_2 has a lower melting point than CaC_2 . [1m]

(iv) While calcium carbide exists as CaC_2 , another ionic compound of Group II metals, beryllium carbide, exists as Be_2C instead. Suggest why this is so.

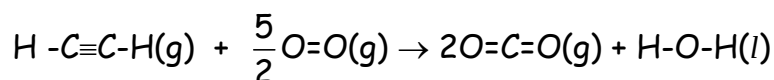
Be^{2+} has a smaller radius and hence a higher charge density than Ca^{2+} . Thus Be^{2+} polarises the C_2^{2-} anion to a larger extent, forming C^{4-} anion. [1m]

(b) Calcium carbide is produced industrially in an electric furnace from a mixture of calcium carbonate and carbon at approximately 2000 °C. It then reacts with water to form ethyne, $H-C\equiv C-H$, as one of the products.

(i) Given that the carbide anion acts like a Bronsted-Lowry base, write an equation of the reaction between calcium carbide and water.



(ii) Ethyne is commonly used for welding metals. With the use of the *Data Booklet*, calculate the enthalpy change of combustion of ethyne. Show your working clearly.

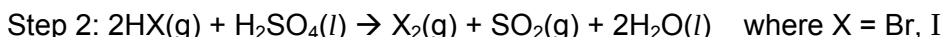
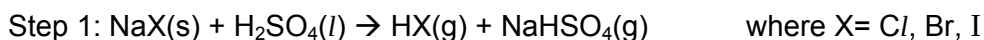


ΔH_c of ethyne

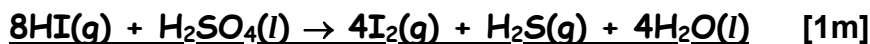
$$= +840 + 2(410) + \frac{5}{2}(496) - 4(740) - 2(460) \quad [1m]$$

$$= \underline{-980 \text{ kJ mol}^{-1}} \quad [1m]$$

- 3 Halogens can be produced when sodium halides react with concentrated sulfuric acid via a two step-reaction with concentrated H_2SO_4 behaving as a strong acid in step 1.



- (a) By means of a balanced equation, show how HI is a stronger reducing agent than HCl and HBr.



A student compared the reactions of sodium halides with other concentrated acids by heating NaCl(s) , NaBr(s) and NaI(s) with concentrated H_3PO_4 , H_2SeO_4 and H_6TeO_4 . The observations from the reactions were summarised in the table below.

	NaCl(s)	NaBr(s)	NaI(s)
H_3PO_4	White fumes evolved		
H_2SeO_4	Yellowish green gas evolved	Brown gas evolved	Violet gas evolved
H_6TeO_4	No observation		

- (b) Based on the observations given above, state and explain which of the three acids H_3PO_4 , H_2SeO_4 or H_6TeO_4 is the

- (i) weakest acid in Step 1.

H_6TeO_4 is the weakest acid [1m] as it is **unable to donate its proton to any of the NaX(s) to form HX(g)** [1m]

- (ii) strongest oxidising agent in Step 2.

H_2SeO_4 is the strongest oxidising agent [1m] as it is **able to oxidise any HX to X_2** . [1m]

- (c) The hydrogen halides can behave as an acid by passing HX gas into water. Explain why HF(aq) is a weak acid while HCl(aq) , HBr(aq) and HI(aq) are strong acids.

Since the **H-F bond has the greatest bond energy** (or the **H-F bond is the shortest bond**), the **H-F bond is the strongest and most difficult to break**, and hence HF is a weak acid that ionises partially. [1m]

The H-Br bond, H-Cl bond and H-I bond are longer and weaker, and hence easier to break. Therefore, HCl(aq) , HBr(aq) and HI(aq) are strong acids which ionises completely.

- 3 (d) Concentrated hydrochloric acid is used together with nitrobenzene and tin in the synthesis of phenylamine as described below.

Preparation of phenylamine

- 1 Place 4 cm³ of nitrobenzene and 10 g of tin in a 250 cm³ two-necked round bottle flask. In one neck is placed a tap funnel and in the other is fitted with a reflux condenser.
- 2 Place 24 cm³ of concentrated hydrochloric acid in the tap funnel and then add the acid a few drops at a time and shake the flask between each addition.
- 3 When all the acid has been added, heat the reaction mixture using a water bath for about 15 minutes.
- 4 Cool the flask and add aqueous sodium hydroxide until the solution is strongly alkaline. Cool the reaction mixture occasionally in ice-water bath.
- 5 Rearrange the apparatus for steam distillation.

- (i) The first step of the synthesis involves the reaction between tin and concentrated hydrochloric acid where tin is converted to tin(II) chloride.

Write an equation to represent this reaction.



- (ii) Explain why the reaction mixture is heated using a water bath (step 3) instead of a bunsen burner.

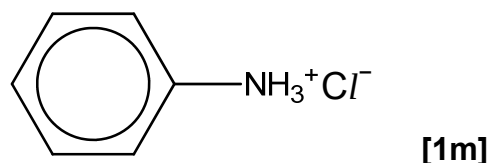
Hydrogen gas produced is highly flammable and can cause explosion in the presence of a naked flame

(or Organic compounds are flammable) [1m]

- (iii) Suggest why the reaction mixture (step 4) has to be cooled upon adding NaOH(aq).

Cooling is required as the acid base reaction of NaOH and HCl evolves large amount of heat. [1m]

- (iv) If NaOH(aq) is **not** added in step 4, another organic compound (instead of phenylamine) will be formed. Give the structural formula of this compound.



3. (d) (v) Phenylamine is removed from the reaction mixture by steam distillation (step 5). In steam distillation, the distilling part is infused with steam, which carries the phenylamine vapour into the condenser, where phenylamine and water is collected as a distillate.

Given that the composition of the distillate can be calculated using the following relationship:

$$\frac{\text{Amount of phenylamine}}{\text{Amount of water}} = \frac{\text{Partial pressure of phenylamine}}{\text{Partial pressure of water}}$$

and that boiling occurs when the sum of the vapour pressure equals to the atmospheric pressure,

calculate the ratio, by mass, of phenylamine to water in the distillate at an atmospheric pressure of 100 kPa when the partial pressure of phenylamine is found to be 8.0 kPa.

$$\text{Partial pressure of water} = 100 - 8 = 92 \text{ kPa}$$

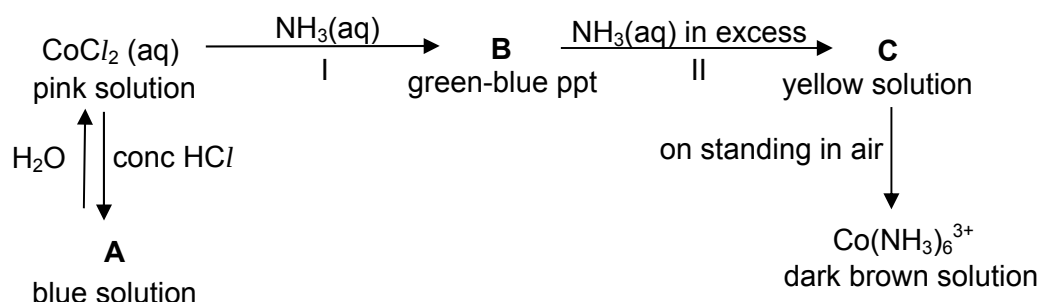
$$\frac{\text{Amount of phenylamine}}{\text{Amount of water}} = \frac{8}{92} \quad [1\text{m}]$$

$$\frac{\text{Mass of phenylamine}}{\text{Mass of water}} = \frac{8 \times 93}{92 \times 18} = \underline{0.449} \quad [1\text{m}] \text{ 3.s.f}$$

4 Use of the *Data Booklet* is relevant to this question.

Cobalt is a typical transition metal; it is hard, fairly unreactive and is used in the production of special alloys. Cobalt also forms many coloured compounds and complexes.

(a) Aqueous cobalt(II) chloride, CoCl_2 , is a pink solution which gives the following reactions.



(i) State the role of NH_3 in reactions I and II.

I: base [1m]

II: ligand [1m]

(ii) State the formula of compound B.

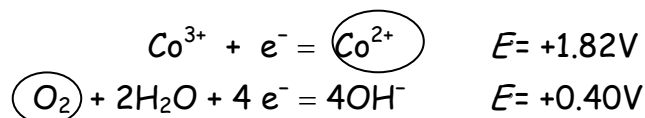
Co(OH)_2 [1m]

(iii) What is the formula of the cation present in $\text{CoCl}_2(\text{aq})$ and the cation present in C?

$\text{CoCl}_2(\text{aq})$: $\text{Co(H}_2\text{O)}_6^{2+}$ [1m]

C: $\text{Co(NH}_3)_6^{2+}$ [1m]

(iv) Aqueous CoCl_2 is chemically stable on standing in air, using relevant E° values, explain why this is so.



$$E_{\text{cell}} = +0.40 - (+1.82) = \underline{-1.42 \text{ V} < 0} \quad [1\text{m}]$$

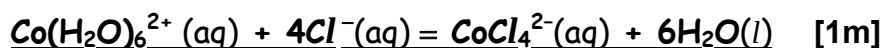
The reaction between $\text{Co}^{2+}(\text{aq})$ and O_2 in the air is energetically not feasible.

(v) When $\text{CoCl}_2(\text{aq})$ is converted into A, one mole of CoCl_2 reacts exactly with two moles of HCl with no other compound formed.

Suggest the formula of the complex ion present in A.

CoCl_4^{2-} [1m]

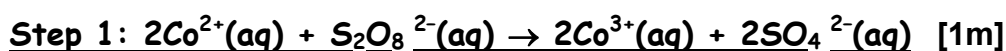
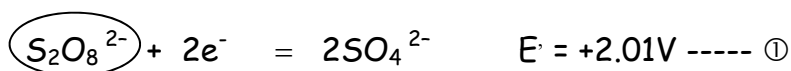
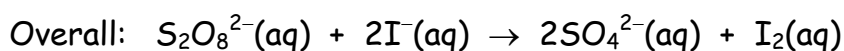
- 4 (a) (vi) Dilution of **A** with water produces the original pink colour. By means of an ionic equation, explain this observation.



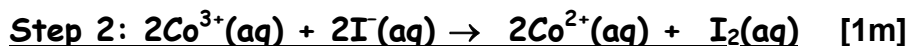
Addition of H_2O shifts the equilibrium position to the left, favouring the formation of pink $\text{Co}(\text{H}_2\text{O})_6^{2+}$. [1m]

- (b) Cobalt and their compounds are useful catalysts. A student proposed using aqueous solution of Co^{2+} to catalyse the reaction between iodide ions, I^- , and peroxodisulfate ions, $\text{S}_2\text{O}_8^{2-}$.

By considering relevant E° values, explain, with the aid of equations, how Co^{2+} ions can catalyse this reaction.



$$E_{\text{cell}} = +2.01 - (+1.82) = \underline{+0.19 \text{ V} > 0 \text{ (energetically feasible)}} \quad [1\text{m}]$$

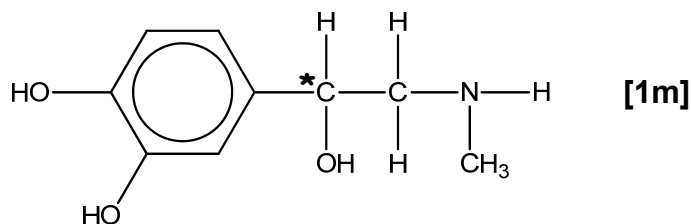


$$E_{\text{cell}} = +1.82 - (+0.54) = \underline{+1.28 \text{ V} > 0 \text{ (energetically feasible)}} \quad [1\text{m}]$$

- (c) When potassium cyanide is added to a solution containing $\text{Co}(\text{NH}_3)_6^{3+}$, a colour change of solution occurs. Suggest an explanation for this observation.

Ligand exchange occurs as the stronger CN^- ligand displaces the weaker NH_3 ligand to form a stronger dative bond with Co^{3+} , hence forming a more stable $\text{Co}(\text{CN})_6^{3-}$ complex. [1m]

- 5 Adrenaline is a hormone which acts as a stimulant when secreted directly into the blood stream. It has the following structure:

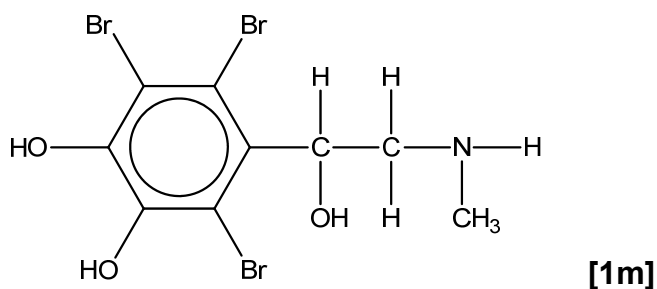


- (a) (i) Name the functional groups present in an adrenaline molecule.

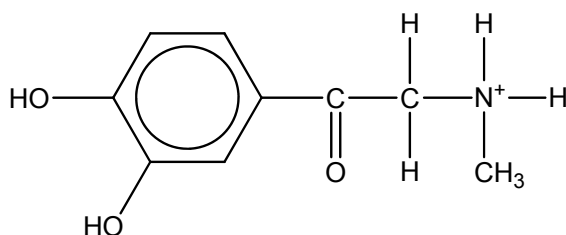
Phenol, secondary alcohol, secondary amine [1m]

- (ii) By means of an asterisk, identify the chiral centre in the structure of adrenaline drawn above.
- (b) Draw the structural formula of **each** of the organic products formed when adrenaline is treated with the following reagents.

- (i) an excess of bromine dissolved in a suitable solvent



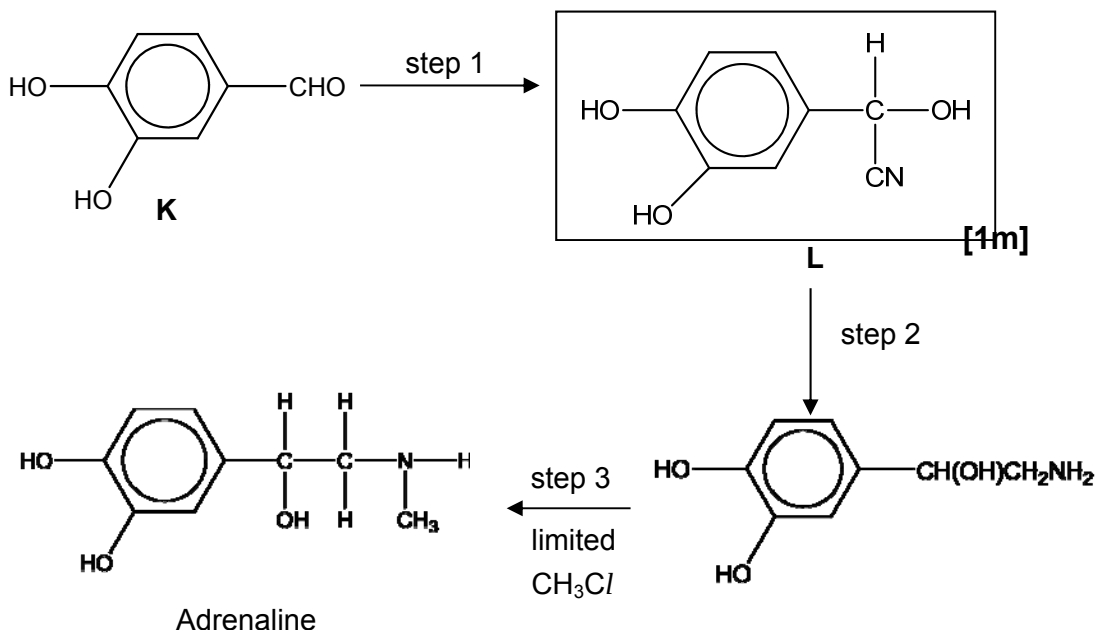
- (ii) hot acidified potassium dichromate(VI)



[1m] Oxidation of alcohol

[1m] Acid-base reaction of amine

5 (c) Adrenaline can be synthesised by the following route.



(i) Draw the structural formula of **L** in the box provided.

(ii) State the reagents and conditions for step 1 and step 2.

Step 1: H₂CN⁻, trace amount of NaCN (or NaOH) [1m]

Step 2: LiAlH₄, dry ether [1m]

(iii) Explain why step 3 is not a good method to synthesise adrenaline.

The yield is poor due to a mixture of products formed as the amine (adrenaline) produced still possesses a lone pair of electrons on its N (or can still act as a nucleophile) which allows further nucleophilic substitution with CH_3Cl . [1m]

(d) State one reagent which will react with **K** and **not** with adrenaline, and describe what will be observed.

Reagent: 2,4-DNPH [1m]

Observation: Orange ppt observed for K but not for adrenaline [1m]

Or, Reagent: Tollens' reagent (or $\text{Ag}(\text{NH}_3)_2^+(\text{aq})$), heat [1m]

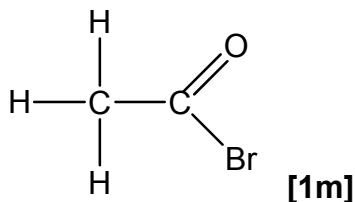
Observation: Silver mirror observed for K but not for adrenaline [1m]

- 6 Esters are used commercially in perfumes and solvents. For example, 2-propyl ethanoate can be made in the laboratory by reacting propan-2-ol with compound **X**. Compound **X** gives a cream precipitate when added to $\text{AgNO}_3(\text{aq})$.

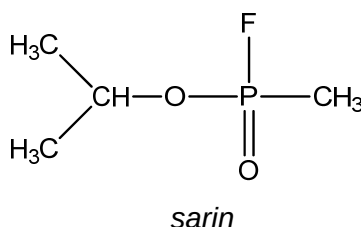
(a) (i) Identify the cream precipitate.

AgBr [1m]

(ii) Draw the displayed formula of compound **X**.



The nerve gas *sarin*, is a phosphate ester which is extremely dangerous as it interferes with the action of a neurotransmitter in the body. It has the following structural formula.



(b) In order to destroy stockpiles of *sarin*, it is necessary to break the substance down by hydrolysis. It reacts in a similar way to an ester.

(i) State the reagents and conditions to break down an ester.

NaOH(aq) (or KOH(aq) or HCl(aq) or H₂SO₄(aq)), heat under reflux
[1m]

(ii) Suggest the structural formulae of the products formed when *sarin* is hydrolysed under similar conditions stated in b(i).

<u>(CH₃)₂CHOH</u> [1m]	[1m] or only for alkaline hydrolysis with NaOH(aq) or KOH(aq)
--	--

- 6 (b) (iii) How would you expect the rate of hydrolysis of *sarin* to differ from that of 2-propyl ethanoate? Explain your answer.

The rate of hydrolysis of sarin is higher/faster. [1m]

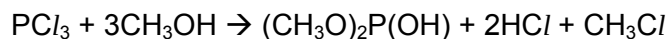
As P atom is bigger than C atom, the P-O bond in sarin is longer and weaker and hence easier to break than the C-O bond in the ester.

Or,

The P atom in sarin is more electron-deficient as it is bonded to one extra highly electronegative F atom (or it is bonded to three highly electronegative atoms, 1 F and 2 O), hence sarin is more susceptible towards nucleophilic attack by H₂O nucleophile than the ester. [1m]

- 6 (c) *Sarin* is made by a series of chemical reactions.

The first stage is to treat methanol with phosphorus trichloride.



By-products produced from various side reactions can reduce the yield of $(\text{CH}_3\text{O})_2\text{P}(\text{OH})$ which is the desired intermediate for the next stage of the synthesis of *sarin*.

- (i) Suggest a phosphorus-containing by-product that could be formed in the same reaction.

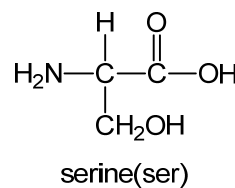
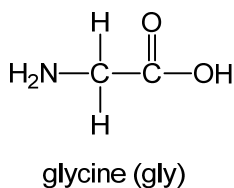
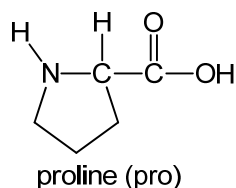


- (ii) Suggest and explain whether NCl_3 is expected to react in a similar manner with methanol.

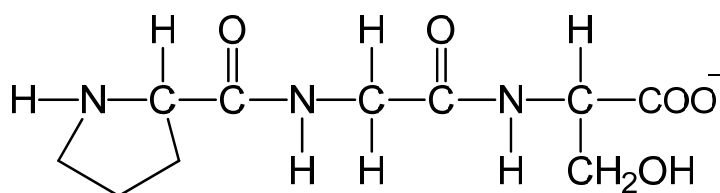
NCl_3 is not expected to react [1m] in a similar manner with methanol.

N in NCl_3 does not have energetically accessible empty d orbital to accept a lone pair of electrons from O atom of CH_3OH molecule to form dative bond with CH_3OH molecule. [1m]

- (d) The formula of three amino acids are given below.



Draw the structure of a tripeptide with the amino acid sequence pro-gly-ser showing the form in which it would exist at pH 12.



[1m] for correct sequence of R group with N terminal on left and C terminal on Right with free $-\text{NH}_2$ and $-\text{COOH}$ ends.

[1m] acid-base reaction $-\text{COOH}$ to $-\text{COO}^-$, ROH remains unchanged

- 6 (e) Pepsin undergoes partial hydrolysis to give the following tripeptides. Using the same 3-letter abbreviations as below, write out the amino acid sequence of the **smallest** polypeptide that could produce the following, given that the free NH_2 (N-terminal) end of the peptide is the amino acid phe.

leu-ser-glu

glu-pro-gly

gly-leu-ser

thr-glu-pro

pro-gly-ser

phe-gly-leu

phe-gly-leu-ser-glu-thr-glu-pro-gly-ser [1m]

- (f) Pepsin is an enzyme which aids digestion in the presence of HCl in the stomach. With an empty stomach, the presence of HCl causes gastric discomfort which can be commonly treated by medicine like *Gaviscon*[®] that contains $\text{Mg}(\text{OH})_2$ as its active ingredient. However, overdosage of *Gaviscon*[®] may result in indigestion. Suggest and explain the impact of overdosage of *Gaviscon*[®] on the catalytic activity of pepsin.

$\text{Mg}(\text{OH})_2$ reacts with HCl which decreases the concentration of HCl and increases the pH above pH 3.

In the **presence of OH^-** ,

$-\text{COOH}$ becomes $-\text{COO}^-$, this disrupts the hydrogen bonds in the tertiary and quaternary structure of pepsin. [1m]

$-\text{NH}_3^+$ becomes $-\text{NH}_2$, this disrupts the ionic interactions in the tertiary and quaternary structure of pepsin.

Presence of Mg^{2+} ions can also form ionic interactions with $-\text{COO}^-$ group in the side-chain of the amino acid residue of the pepsin. This disrupts the ionic interactions in the tertiary and quaternary structures of pepsin. [1m]

As a result, the **conformation (or 3D shape) of the active site of pepsin is altered** and the **catalytic activity of the pepsin is inhibited** since the **substrate can no longer bind to the active site.** [1m]